



Addis Ababa Science and Technology University
Department of Electromechanical Engineering

Internship Project

Submitted by:

1. [Full Name] ID: [Student ID]
 2. [Full Name] ID: [Student ID]
 3. [Full Name] ID: [Student ID] *(Delete unused lines for individual*
 4. [Full Name] ID: [Student ID]
 5. [Full Name] ID: [Student ID]
- submission)*

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Abstract

This report presents the design, modeling, simulation, and implementation of a motor-driven conveyor belt system tailored for Industry [YZ]. Utilizing a 24V DC geared motor, IR proximity sensors, and Arduino-based control, the system was systematically modeled in MATLAB/Simulink. Simulation results guided mechanical and electrical design decisions. The final prototype demonstrated a

15

Keywords: Conveyor system, DC motor, Modeling, Simulation, LQR control, Arduino, Mechatronics

Contents

1	Introduction	5
1.1	Background of the Study	5
1.2	Problem Statement	5
1.3	Objectives	5
1.4	Scope and Limitation	5
1.5	Methodology Overview	6
2	Literature Review	7
2.1	Overview of Conveyor Systems	7
2.2	Motor Drive Techniques	7
2.3	Sensing and Control	7
2.4	Simulation in Mechatronics	7
3	System Design	8
3.1	Mechanical Design	8
3.1.1	Load Analysis	8
3.1.2	Design of Machine Elements	8
3.1.3	Belt and Pulley Design	8
3.1.4	Bearing Design	8
3.1.5	Shaft Design	8
3.1.6	Fabrication and Assembly	8
3.2	Mathematical Modeling	9
3.2.1	Modeling of Single Inverted Pendulum	9
3.2.2	Linear State-Space Representation of IP System	9
3.2.3	Linearization Using Taylor Expansion & Jacobian	9
3.3	Controller Design	9
3.3.1	Open-loop Response Analysis	9
3.3.2	Controllability and Observability	9
3.3.3	Control Strategy Selection	9
3.3.4	LQR Controller Implementation	10
3.3.5	Control System Architecture	10
3.4	Electrical and Electronic System Design	10

4	Simulation and Prototype/Manufacturing Results and Discussion	11
4.1	Simulation Results and Analysis	11
4.2	Prototype/Experimentation Results and Analysis	12
4.3	Discussion (Instructions)	13
5	Conclusion and Recommendation	14
5.1	Conclusion	14
5.2	Recommendations	14

List of Figures

1	Simulink block diagram of conveyor system	11
2	Time(in seconds) Vs. Angular Displacement	12
3	Time(in seconds) Vs. Angular Displacement	13
4	CAD view of conveyor belt system	15

List of Tables

1	System Performance Comparison	13
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1 Introduction

1.1 Background of the Study

Automated conveyor systems are integral to industrial operations, enhancing efficiency, reliability, and throughput. Industry [YZ] experienced issues with misalignment, energy inefficiency, and high maintenance. This motivated a comprehensive redesign using mechatronic modeling and simulation techniques.

1.2 Problem Statement

The existing system in Industry [YZ] exhibited:

- Misalignment rate of 8%
- Energy consumption of 120W/m
- Weekly maintenance of 3hr

1.3 Objectives

General Objective: Develop an efficient, accurate, and reliable conveyor system through design, modeling, simulation, and implementation.

Specific Objectives:

1. Analyze and model system dynamics
2. Simulate control strategies
3. Design mechanical and electrical components
4. Implement Arduino-based control
5. Evaluate system performance

1.4 Scope and Limitation

Focus is on low-to-medium duty conveyor systems with DC motor control; PLC integration and pneumatic systems are excluded.

1.5 Methodology Overview

1. System analysis of current setup
2. Mathematical modeling
3. Simulation in MATLAB/Simulink
4. Mechanical/electrical design
5. Prototyping and testing

2 Literature Review

2.1 Overview of Conveyor Systems

Discusses relevant literature on conveyor types, belt materials, and industrial applications.

2.2 Motor Drive Techniques

DC motors with PWM control are widely employed for conveyor applications due to variable speed control and torque.

2.3 Sensing and Control

IR sensors, encoders, and closed-loop control strategies enhance conveyor accuracy and adaptability.

2.4 Simulation in Mechatronics

Advancements in model-based design using MATLAB/Simulink and Simscape facilitate early validation.

3 System Design

3.1 Mechanical Design

3.1.1 Load Analysis

Total load calculation:

$$F_{total} = F_{belt} + F_{rollers} + F_{payload}$$

3.1.2 Design of Machine Elements

Fasteners, frame supports, and guards designed per standard mechanical design procedures, validated by FEA.

3.1.3 Belt and Pulley Design

Belt speed relation:

$$V = \frac{\pi DN}{60}$$

3.1.4 Bearing Design

L10 life for bearings:

$$L_{10} = \left(\frac{C}{F}\right)^3 \times 10^6$$

3.1.5 Shaft Design

Shaft shear stress formula:

$$\tau_{max} = \frac{16T}{\pi d^3}$$

3.1.6 Fabrication and Assembly

Fabrication methods include CNC and machining of frame, pulleys, shafts; detailed alignment procedures are documented.

3.2 Mathematical Modeling

3.2.1 Modeling of Single Inverted Pendulum

Euler–Lagrange method applied to derive dynamics:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = Q_i$$

3.2.2 Linear State-Space Representation of IP System

Define:

$$x = [\theta, \dot{\theta}, x, \dot{x}]^T$$

State-space model:

$$\dot{x} = Ax + Bu, \quad y = Cx + Du$$

3.2.3 Linearization Using Taylor Expansion & Jacobian

Linearized system around equilibrium:

$$A = \left. \frac{\partial f}{\partial x} \right|_{x_0, u_0}, \quad B = \left. \frac{\partial f}{\partial u} \right|_{x_0, u_0}$$

3.3 Controller Design

3.3.1 Open-loop Response Analysis

Analysis via step response, root locus, pole-zero to evaluate natural dynamics.

3.3.2 Controllability and Observability

Check with Kalman matrices:

$$\text{rank}[B \ AB \ A^2B \ \dots] = n$$

3.3.3 Control Strategy Selection

LQR selected for optimal performance-weight tradeoff.

3.3.4 LQR Controller Implementation

Cost function:

$$J = \int_0^{\infty} (x^T Q x + u^T R u) dt$$

Optimal gain:

$$K = R^{-1} B^T P$$

3.3.5 Control System Architecture

Block diagram integrates sensor feedback, LQR control, motor driver, and actuator.

3.4 Electrical and Electronic System Design

Key components:

- 24V DC geared motor
- IR proximity sensors
- L298N motor driver
- Arduino Uno
- Power regulation and protection

4 Simulation and Prototype/Manufacturing Results and Discussion

4.1 Simulation Results and Analysis

Provide a detailed analysis of your simulation output and prototype testing. Include diagrams, tables, plots, and images to compare simulation and real-world behavior.

Example: The conveyor belt and LQR control system were modeled in Simulink using DC motor dynamics, IR sensor simulation, and feedback control loops.

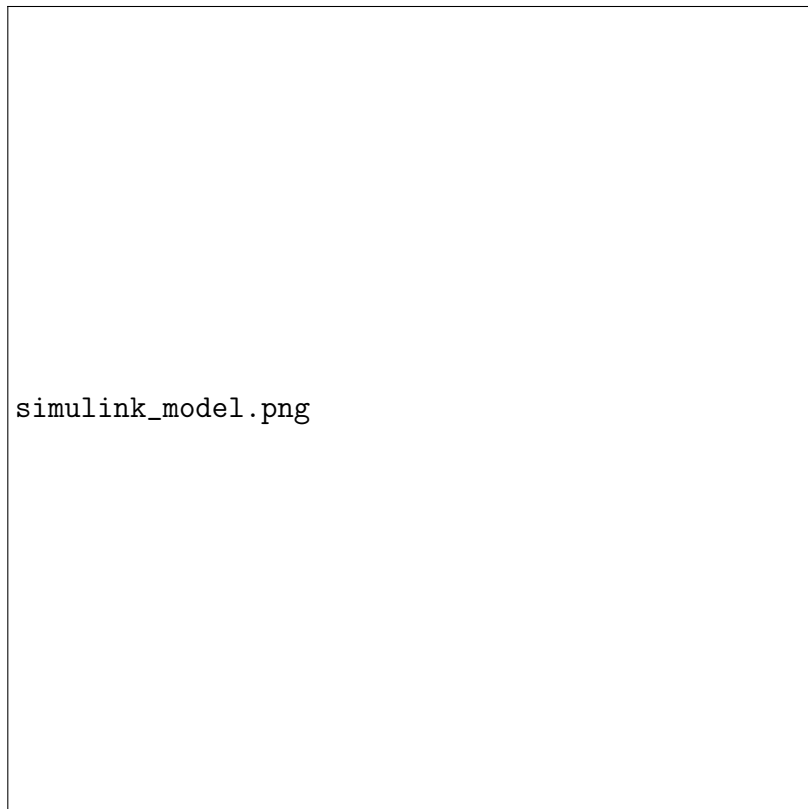


Figure 1: Simulink block diagram of conveyor system

The simulation results indicate that the system achieved a rise time of 1.2 seconds, with a steady-state error of less than 1% and an overshoot of 3.5%. These values suggest a well-tuned control response, balancing speed and accuracy effectively for the conveyor belt system.

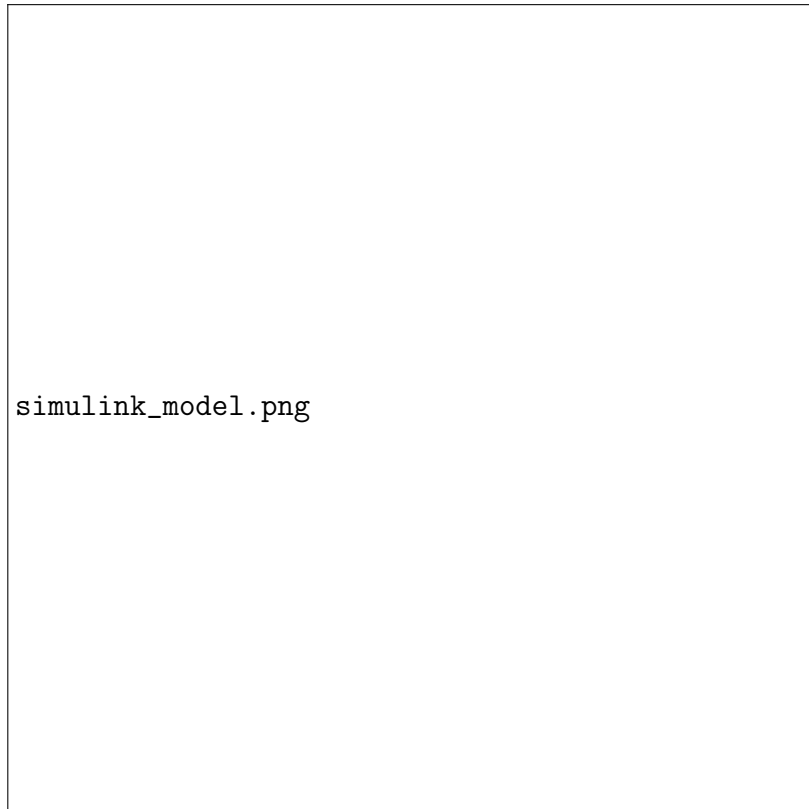


Figure 2: Time(in seconds) Vs. Angular Displacement

4.2 Prototype/Experimentation Results and Analysis

Explain your physical setup, hardware components, and safety measures. Include:

- Photos of the prototype
- List of equipment and specifications
- Test procedures



Figure 3: Time(in seconds) Vs. Angular Displacement

Prototype Results (Example)

Table 1: System Performance Comparison

Metric	Simulated	Prototype
Energy Usage (W/m)	105	102
Position Accuracy (%)	96	98
Maintenance Time (hr/week)	N/A	1

4.3 Discussion (Instructions)

Interpret and compare simulation and prototype results. Address discrepancies, modeling limitations, hardware non-linearities, and environmental effects.

5 Conclusion and Recommendation

5.1 Conclusion

Summarize your overall findings, system performance, and whether objectives were met. Discuss areas of success and lessons learned.

Example: The project successfully integrated modeling, simulation, and physical implementation. The LQR-controlled conveyor system met the design targets for accuracy and energy efficiency.

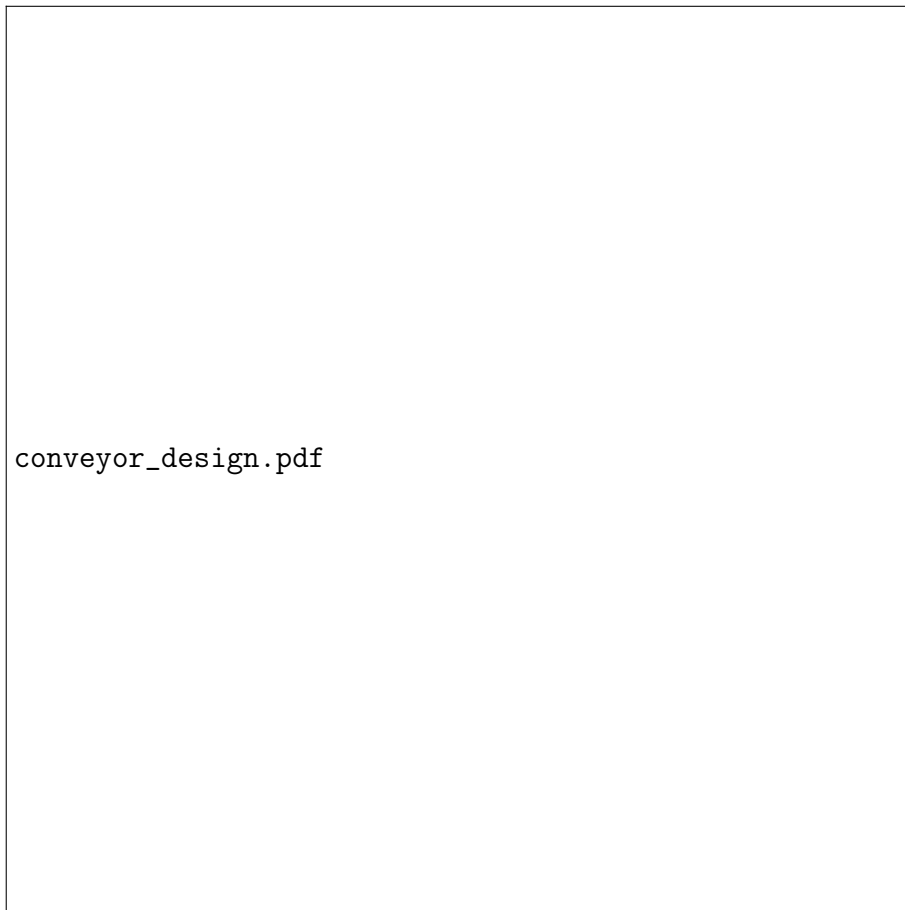
5.2 Recommendations

Example:

- Use of encoder feedback for precise speed control
- Upgrade to industrial-grade components
- Incorporate IoT for remote monitoring
- Explore fuzzy/PID hybrid control for robustness

Appendices

Appendix A: CAD Drawings



conveyor_design.pdf

Figure 4: CAD view of conveyor belt system

Appendix B: Arduino Code

References

Please ensure all cited sources in your report are listed here using APA 7th edition style. Include journal articles, books, websites, standards, or any documentation used in modeling, simulation, or implementation.